

Air University



Organization & Assembly (Lab)

Year 2025

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Assignment # 02

Corrected Code:

```
ASM Question01.asm > ...
1  .MODEL SMALL
2  .STACK 100h
3
4  .DATA
5      A    DW 2    ; Define A as a word (16-bit)
6      B    DW 5    ; Define B as a word (16-bit)
7      SUM  DW ?    ; Uninitialized word for sum
8
9  .CODE
10 MAIN PROC
11     ; Initialize data segment
12     MOV AX, @DATA
13     MOV DS, AX
14
15     ; Add the numbers
16     MOV AX, A      ; Load A into AX
17     ADD AX, B      ; Add B to AX (Result is now in AX)
18
19     ; Prepare to display the result
20     MOV AL, AL      ; Ensure we're working with the low byte
21     ADD AL, 30h     ; Convert low byte to ASCII
22
23     MOV DL, AL      ; Move to DL for output
24     MOV AH, 02h     ; Output character
25     INT 21h        ; Call DOS interrupt to display
26
27     ; Exit to DOS
28     MOV AX, 4C00h   ; Terminate with exit code 0
29     INT 21h        ; Call DOS interrupt
30
31 MAIN ENDP
32 END MAIN          ; End of program
```

Here are the changes made to fix the program's issues:

- **Corrected Typos and Syntax Errors:** We fixed misspelled commands (for example, changed `.STACK !OOH` to `.STACK 100h`), revised variable definitions (like changing `ow 2` to `A DW 2`), and corrected instructions (for instance, `AD!:> AX,B` became `ADD AX, B`).
- **Fixed Procedure Structure Problems:** We changed the procedure ending from `MAIN END` to `MAIN ENDP` to follow standard assembly rules (like TASM/MASM). This change removed the "Open procedure" warning and the "Illegal instruction" error.
- **Simplified Code Logic:** We eliminated unnecessary memory accesses. For example, we used `AX` directly for the sum instead of storing and loading from `SUM` repeatedly. This made the code simpler and helped with the `BYTE PTR` issue, improving overall compatibility.
- **Enhanced Overall Structure:** We added comments to increase clarity, ensured proper segment setup, and refined the addition and output logic to prevent errors during assembly.

Output

```
Z:\>mount d ./code
Drive D is mounted as local directory ./code/

Z:\>d:

D:\>set PATH=C:\TASM

D:\>TASM D:\test.asm
Turbo Assembler Version 4.1 Copyright (c) 1988, 1996 Borland International

Assembling file:   D:\test.asm to test.OBJ
Error messages:    None
Warning messages:  None
Passes:            1
Remaining memory:  468k

D:\>TLINK D:\test
Turbo Link Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International

D:\>D:\test
7
D:\>{
```

THE END
